



What is Kubernetes?



- Open-source system to manage and run containers at scale
- Automatically handles deployment, scaling, and healing of apps
- Runs your containers across multiple machines (nodes)
- Restarts containers if they crash
- Distributes traffic between app instances (load balancing)





What is GKE?

(Google Kubernetes Engine)



- Google Cloud ka managed Kubernetes service
- Google runs & manages the cluster infrastructure for you
- No need to manage master nodes manually
- One-click cluster creation from console
- Built-in auto scaling, auto repair, monitoring
- Direct integration with other GCP services



Why GKE is Considered the Main Power of GCP



- Kubernetes originally inspired by Google's internal Borg system
- Google has deepest real-world container experience
- Very strong DevOps + CI/CD + autoscaling ecosystem
- GKE Autopilot = near-serverless Kubernetes experience
- Used in production by high-scale, modern platforms

